Trinity Engine Canon Extract

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1 Overview

- Trinity Engine v1 is a deterministic, stateless calculation engine.
- Governing implementation canon: Documents 01-08 and 11.
- Test canon: Document 10.
- Control and transparency documents: Document 00, Document 09, Document 12.
- Canon precedence order:
  - Document 04 Input Specification
  - Document 05 Calculation Canon
  - Document 06 Mapping Canon

  - Document 07 Output Specification
  - Document 03 Canonical Source Stack
  - Document 02 Supported Objects and Scope
  - Document 01 Purpose and System Boundary
  - Document 09 Ambiguity Register
  - Document 00 Canon Build Status

2 Package State

- Document 00: Draft control document.
- Document 01: Near Final.
- Document 02: Near Final.
- Document 03: Near Final.
- Document 04: Final.
- Document 05: Near Final.
- Document 06: Near Final.
- Document 07: Near Final.
- Document 08: Near Final.
- Document 09: Draft Approved, actively in use.
- Document 10: Draft Approved, awaiting fixture population.
- Document 11: Near Final.
- Major active package risk recorded in the canon itself: final consistency pass across Documents 05, 06, and 07.

3 Scope And Boundary

- In scope:
  - input validation against canonical normalized input
  - astrology calculations

  - Human Design calculations
  - Gene Keys derivation
  - application of pinned mappings, thresholds, ordering, and formatting
  - structured success and error responses
  - deterministic metadata for reproducibility
- Out of scope:
  - birthplace search, geocoding, and timezone determination
  - UI and presentation behavior
  - persistence and operational logging
  - narrative, content, recommendations, coaching, and translation
  - any calculation or output not explicitly enumerated in the canon
- Supported domains:
  - Astrology
  - Human Design
  - Gene Keys
- No separate "Trinity derived calculations" category exists in v1.
- Engine boundary rule:
  - input reaches the engine only after normalization
  - raw birthplace strings, unresolved timezone data, free-text dates/times, and ambiguous location data are outside scope

4 Authoritative Sources And Version Pins

- Swiss Ephemeris is the authoritative astronomical source.
- Swiss Ephemeris version pin: `2.10.03`.
- IANA Time Zone Database is the authoritative timezone standard.
- Gregorian calendar is the authoritative civil calendar basis.
- Tropical zodiac is authoritative across domains.

- Placidus is the authoritative astrology house system.
- Node policy by domain:
  - Astrology uses Mean Node.
  - Human Design uses True Node.
  - Gene Keys inherits the Human Design basis.
- External libraries are implementation vehicles only; library defaults are non-authoritative unless canon explicitly adopts them.
- Third-party platforms may be used only for sanity checking, not as truth.

5 Supported Objects And Outputs

5.1 Astrology

- Raw objects:
  - sun
  - moon
  - mercury
  - venus
  - mars
  - jupiter
  - saturn
  - uranus
  - neptune
  - pluto
  - chiron
  - $\text{north}_{\text{nodemean}}$
  - earth
- Additional supported outputs:
  - ascendant
  - midheaven

  - house cusps 1-12

- Per-object supported attributes:
  - $\text{object}_{\text{id}}$
  - longitude
  - sign
  - house

- Explicitly out of scope:
  - aspects
  - dignities
  - dispositors
  - house rulers
  - elemental or modality synthesis
  - asteroids beyond Chiron
  - fixed stars
  - interpretive text

5.2 Human Design

- Two snapshots:
  - personality at birth
  - design at $\text{design}_{\text{timeutc}}$

- Snapshot object identifiers and order:
  1. sun
  2. earth
  3. $\text{north}_{\text{node}}$
  4. $\text{south}_{\text{node}}$
  5. moon
  6. mercury
  7. venus
  8. mars

9. jupiter
10. saturn
11. uranus
12. neptune
13. pluto

- Activation output for each snapshot object:
  - $\text{object}_{\text{id}}$
  - gate
  - line
- Structural outputs:
  - channels
  - centers
  - definition
  - type
  - authority
  - profile
  - $\text{incarnation}_{\text{cross}}$

5.3 Gene Keys

- Not an independent astronomical system.
- Derived strictly from Human Design gate and line outputs.
- Supported positions:
  - $\text{life}_{\text{work}}$ = Personality Sun
  - evolution = Personality Earth
  - radiance = Design Sun
  - purpose = Design Earth
- Supported fields per position:
  - key
  - line

- Out of scope:
  - shadow, gift, and essence text
  - sequence prose
  - semantic states

6 Input Contract

6.1 Canonical Payload

- One person per payload.
- Required fields:

field	type	format / rule
$\text{birth}_{\text{date}}$	string	YYYY-MM-DD
$\text{birth}_{\text{time}}$	string	HH:MM, 24-hour, minute only
timezone	string	IANA timezone identifier
latitude	number	decimal degrees, -90.0..90.0
longitude	number	decimal degrees, -180.0..180.0

- Canonical logical shape:

```
{
  "birth_date": "YYYY-MM-DD",
  "birth_time": "HH:MM",
  "timezone": "Area/Location",
  "latitude": 0.000000,
  "longitude": 0.000000
}
```

6.2 Validation Rules

- Validate before any calculation:
  - presence of all required fields
  - exact field format compliance
  - type compliance
  - numeric range compliance
  - IANA-compliant timezone format

  - Gregorian date validity
  - 24-hour minute time validity

- Strict numeric typing:
  - numeric fields must be numeric values, not strings
  - no silent coercion of numeric-looking strings
- No optional canonical input fields in v1.
- Extra upstream fields are non-authoritative and must not affect results.

6.3 Normalization Assumptions

- Before engine submission:
  - location already resolved to unique latitude and longitude
  - timezone already resolved to a valid IANA identifier
  - date already resolved to Gregorian YYYY-MM-DD
  - time already normalized to HH:MM
  - ambiguity, alias handling, and UI cleaning already completed upstream
  - payload already reduced to one birth event

6.4 Rejection Rules

- `incomplete_input`
  - missing one or more required fields
- `invalid_input`
  - malformed date or time
  - invalid timezone string
  - out-of-range latitude or longitude
  - numeric field supplied as string
- `unsupported_input`
  - structurally valid input outside supported v1 scope

6.5 No Silent Repair

- The engine must not silently:
  - infer missing fields
  - correct malformed values
  - coerce ambiguous inputs
  - map timezone aliases
  - resolve place names
  - truncate unexpected time formats
  - substitute defaults

7 Calculation Canon

7.1 Shared Pre-Calculation Steps

- Validate input.
- Construct local birth datetime from date, time, timezone.
- Convert to UTC using the pinned timezone standard.
- Derive Julian Day or equivalent astronomical time input.
- Use provided latitude and longitude as authoritative coordinates.

7.2 Angular Mechanics

- Canonical longitude range: `[0.0, 360.0)`.
- Normalize by modular arithmetic.
- `360.0` normalizes to `0.0`.
- Opposition derivation uses canonical 180-degree opposition logic.
- Cyclic interval handling must be deterministic.

7.3 Astrology

- Query or derive longitudes at the birth moment.
- Earth longitude = Sun longitude + 180 degrees modulo 360.
- Sign placement is derived from the tropical zodiac intervals.
- House calculation uses:
  - birth moment
  - latitude
  - longitude
  - Placidus
- Return:
  - ascendant
  - midheaven
  - house cusps 1-12
  - supported astrology objects in canonical order

7.4 Human Design Design-Time Derivation

- Design time is a second astronomical moment before birth.
- Design-time target rule:
  - find the prior moment where Sun longitude is exactly 88.0 degrees earlier than at birth
  - target longitude = normalize($\text{birth}_{\text{sunlongitude}}$ - 88.0)
- Search direction:
  - backward in time only
- Initial bracket:
  - centered around approximately 88 days before birth
- Root-finding method:
  - iterative bisection

- Stop conditions:
  - absolute Sun-longitude difference from target < 0.0001 degrees
  - or remaining interval width < 1 second
- Forbidden shortcuts:
  - fixed 88-day subtraction
  - fixed 3-month subtraction
  - rough platform imitation
  - average ephemeris approximation
  - precomputed lookup shortcut

7.5 Human Design Gate And Line Mapping

- Mandala parameters:
  - anchor defined in Document 06
  - gate width = 5.625 degrees
  - line width = 0.9375 degrees
  - direction = ascending ecliptic longitude
  - intervals are start-inclusive and end-exclusive
  - gate order is a fixed lookup list
- Relative mandala position:
  - normalize longitude
  - subtract canonical anchor
  - normalize result into `[0.0, 360.0)`
- Gate assignment:
  - $\text{floor}(\text{relative}_{\text{position}} \,/\, 5.625)$
  - use result as zero-based index into canonical gate order
- Line assignment:
  - compute offset inside gate segment
  - floor(offset / 0.9375)

  - convert to line number 1..6

- Boundary rule:
  - an interval end belongs to the next segment

- Gene Keys uses exactly the same gate and line results as Human Design.

7.6 Canonical Human Design Gate Order

- Gate 38 begins at `277.5°`.

- Gate 41 begins at `300.0°`.

- Canonical 64-gate sequence:
  - [38, 54, 61, 60, 41, 19, 13, 49, 30, 55, 37, 63, 22, 36, 25, 17, 21, 51, 42, 3, 27, 24, 2, 23, 8, 20, 16, 35, 45, 12, 15, 52, 39, 53, 62, 56, 31, 33, 7, 4, 29, 59, 40, 64, 47, 6, 46, 18, 48, 57, 32, 50, 28, 44, 1, 43, 14, 34, 9, 5, 26, 11, 10, 58]

7.7 Human Design Structural Derivation

- Combined activation set:
  - all valid personality activations
  - all valid design activations

- Channels:
  - active only when both gates of a canonical pair are active
  - each active channel returned once

- Centers:
  - defined if they participate in at least one active canonical channel
  - otherwise undefined

- Definition classes:
  - none
  - single
  - split

  - $\text{triple}_{\text{split}}$
  - $\text{quadruple}_{\text{split}}$

- Type derivation:
  - reflector if definition = none
  - if sacral defined:
    * $\text{manifesting}_{\text{generator}}$ if at least one motor-to-throat path exists
    * generator otherwise
  - if sacral undefined and a non-sacral motor connects to throat:
    * manifestor
  - projector otherwise

- Authority derivation priority:
  1. emotional if $\text{solar}_{\text{plexus}}$ defined
  2. sacral if type is generator or $\text{manifesting}_{\text{generator}}$
  3. splenic if spleen defined
  4. $\text{ego}_{\text{manifested}}$ if type is manifestor and ego defined
  5. $\text{ego}_{\text{projected}}$ if type is projector and ego defined
  6. $\text{self}_{\text{projected}}$ if type is projector and g defined
  7. mental if type is projector
  8. lunar if type is reflector

- Profile:
  - `personality_sun_line/design_sun_line`

- Incarnation Cross:
  - structural encoding of $\text{personality}_{\text{sun}}$, $\text{personality}_{\text{earth}}$, $\text{design}_{\text{sun}}$, $\text{design}_{\text{earth}}$
  - no human-readable cross name required

7.8 Gene Keys

- $\text{life}_{\text{work}}$ = Personality Sun
- evolution = Personality Earth
- radiance = Design Sun
- purpose = Design Earth
- Key number equals Human Design gate number.
- Line equals Human Design line number.

8 Mapping Canon

8.1 Astrology Mappings

- Sign order:
  - aries, taurus, gemini, cancer, leo, virgo, libra, scorpio, sagittarius, capricorn, aquarius, pisces
- Sign interval rule:
  - each sign spans 30 degrees
  - interval start inclusive, end exclusive
- Astrology object identifiers:
  - sun
  - moon
  - mercury
  - venus
  - mars
  - jupiter
  - saturn
  - uranus
  - neptune
  - pluto
  - chiron

  - $\text{north}_{\text{nodemean}}$
  - earth

- House identifiers:
  - integers 1..12

8.2 Human Design Centers

- Canonical center list and order:
  1. head
  2. ajna
  3. throat
  4. g
  5. ego
  6. $\text{solar}_{\text{plexus}}$
  7. sacral
  8. spleen
  9. root

- Motor centers:
  - root
  - sacral
  - $\text{solar}_{\text{plexus}}$
  - ego

8.3 Canonical Channel Table

$channel_{id}$	$gate_a$	$gate_b$	$center_a$	$center_b$
1-8	1	8	g	throat
2-14	2	14	g	sacral
3-60	3	60	sacral	root
4-63	4	63	ajna	head
5-15	5	15	sacral	g
6-59	6	59	$solar_{plexus}$	sacral
7-31	7	31	g	throat
9-52	9	52	sacral	root
10-20	10	20	g	throat
10-34	10	34	g	sacral
10-57	10	57	g	spleen
11-56	11	56	ajna	throat
12-22	12	22	throat	$solar_{plexus}$
13-33	13	33	g	throat
16-48	16	48	throat	spleen
17-62	17	62	ajna	throat
18-58	18	58	spleen	root
19-49	19	49	root	$solar_{plexus}$
20-34	20	34	throat	sacral
20-57	20	57	throat	spleen
21-45	21	45	ego	throat
23-43	23	43	throat	ajna
24-61	24	61	ajna	head
25-51	25	51	g	ego
26-44	26	44	ego	spleen
27-50	27	50	sacral	spleen
28-38	28	38	spleen	root
29-46	29	46	sacral	g
30-41	30	41	$solar_{plexus}$	root
32-54	32	54	spleen	root
34-57	34	57	sacral	spleen
35-36	35	36	throat	$solar_{plexus}$
37-40	37	40	$solar_{plexus}$	ego
39-55	39	55	root	$solar_{plexus}$
42-53	42	53	sacral	root
47-64	47	64	ajna	head

8.4 Identifier Normalization

- Canonical case:
  - lowercase $\text{snake}_{\text{case}}$ for named identifiers
  - integers for gate and line values
  - hyphen-joined ascending gate pairs for channel identifiers
- No alias drift:
  - `ego` is canonical
  - `will` must not appear as a separate engine identifier

9 Output Contract

9.1 Success Response

- Top-level fields:
  - status
  - metadata
  - $\text{input}_{\text{echo}}$
  - astrology
  - $\text{human}_{\text{design}}$
  - $\text{gene}_{\text{keys}}$
- Success status:
  - `success`

9.2 Metadata

- Required string fields:
  - $\text{engine}_{\text{version}}$
  - $\text{canon}_{\text{version}}$
  - $\text{source}_{\text{stack}\,\text{version}}$
  - $\text{input}_{\text{schema}\,\text{version}}$
  - $\text{mapping}_{\text{version}}$

- Metadata must not include:
  - random run ids
  - environment names
  - operational log ids
  - timestamps unless later added by canon revision

9.3 Input Echo

- Required fields:
  - $\text{birth}_{\text{date}}$
  - $\text{birth}_{\text{time}}$
  - timezone
  - latitude
  - longitude
- Input echo must use canonical formats and must not include excluded intake fields.

9.4 Astrology Output

- Required sections:
  - system
  - angles
  - $\text{house}_{\text{cusps}}$
  - objects

- Astrology system values:
  - zodiac = `tropical`
  - $\text{house}_{\text{system}}$ = `placidus`
  - $\text{node}_{\text{type}}$ = `mean`
- Angles:
  - ascendant {longitude, sign}
  - midheaven {longitude, sign}

- House cusps:
  - ordered houses 1..12
  - each cusp {house, longitude, sign}
- Astrology object order:
  1. sun
  2. moon
  3. mercury
  4. venus
  5. mars
  6. jupiter
  7. saturn
  8. uranus
  9. neptune
  10. pluto
  11. chiron
  12. $\text{north}_{\text{nodemean}}$
  13. earth

9.5 Human Design Output

- Required sections:
  - system
  - $\text{personality}_{\text{activations}}$
  - $\text{design}_{\text{activations}}$
  - channels
  - centers
  - definition
  - type
  - authority
  - profile
  - $\text{incarnation}_{\text{cross}}$

- Human Design system:
  - $\text{node}_{\text{type}}$ = `true`
  - $\text{design}_{\text{timeutc}}$
- Channels:
  - ordered lexicographically by `channel_id`
  - each channel $\{\text{channel}_{\text{id}}, \text{gate}_{\text{a}}, \text{gate}_{\text{b}}, \text{center}_{\text{a}}, \text{center}_{\text{b}}\}$
- Centers:
  - ordered as head, ajna, throat, g, ego, $\text{solar}_{\text{plexus}}$, sacral, spleen, root
  - each center $\{\text{center}_{\text{id}}, \text{state}\}$
- Allowed definition values:
  - none
  - single
  - split
  - $\text{triple}_{\text{split}}$
  - $\text{quadruple}_{\text{split}}$
- Allowed type values:
  - reflector
  - generator
  - $\text{manifesting}_{\text{generator}}$
  - manifestor
  - projector
- Allowed authority values:
  - emotional
  - sacral
  - splenic
  - $\text{ego}_{\text{manifested}}$
  - $\text{ego}_{\text{projected}}$
  - $\text{self}_{\text{projected}}$
  - mental
  - lunar

9.6 Gene Keys Output

- Required sections:
  - system
  - activations
- Gene Keys system:
  - $\text{derivation}_{\text{basis}} = \texttt{human_design}$
- Activation keys:
  - $\text{life}_{\text{work}}$
  - evolution
  - radiance
  - purpose
- Each activation = {key, line}=.

9.7 Error Output

- Top-level fields:
  - status
  - metadata
  - error
- Error status:
  - `error`
- Error object:
  - $\text{error}_{\text{type}}$
  - message
- Allowed error types:
  - $\text{invalid}_{\text{input}}$
  - $\text{incomplete}_{\text{input}}$
  - $\text{unsupported}_{\text{input}}$
  - $\text{canon}_{\text{conflict}}$
  - $\text{execution}_{\text{failure}}$

9.8 Formatting Rules

- Longitude fields:
  - numeric JSON values
  - rounded to 6 decimal places
- `design_time_utc`:
  - ISO 8601 UTC with trailing `Z`
  - whole-second precision

10 Determinism And Versioning

- Reproducibility requires identical:
  - normalized input
  - $\text{canon}_{\text{version}}$
  - $\text{source}_{\text{stackversion}}$
  - $\text{mapping}_{\text{version}}$
  - $\text{input}_{\text{schemaversion}}$
  - $\text{engine}_{\text{version}}$ where behavior depends on implementation
- No hidden state:
  - no mutable cache state
  - no implicit environment defaults
  - no third-party service responses
  - no stored run history
- Any behavioral change requires:
  - canon review
  - canon revision where applicable
  - regression updates where applicable
  - version impact decision
- Version changes:

- $\text{canon}_{\text{version}}$ changes for scope, calculation, mapping, output, precedence, formatting, or input behavior changes
- $\text{input}_{\text{schemaversion}}$ changes for field/format/type/validation changes
- $\text{mapping}_{\text{version}}$ changes for gate order, channel table, center list, channel-to-center mapping, identifier schemes, or deterministic mapping changes
- $\text{source}_{\text{stackversion}}$ changes for authoritative external source changes
- $\text{engine}_{\text{version}}$ changes for any production implementation change

11 Golden Test Pack Requirements

- Minimum active cases:
  - $\text{valid}_{\text{baseline}}$: 3
  - $\text{valid}_{\text{edge}}$: 5
  - $\text{invalid}_{\text{input}}$: 5
  - $\text{incomplete}_{\text{input}}$: 5
  - $\text{unsupported}_{\text{input}}$: 2
  - $\text{regression}_{\text{sentinel}}$: 3
- Baseline coverage must include:
  - accepted normalized input
  - astrology output
  - Human Design output
  - Gene Keys output
  - metadata
  - $\text{input}_{\text{echo}}$
  - ordering
  - formatting
- Edge coverage must include:
  - sign boundaries
  - house boundaries where testable
  - gate boundaries

  - line boundaries
  - longitude normalization
  - Earth opposition
  - South Node opposition
  - design-time target precision behavior
- Regression sentinels must include at least:
  - strict numeric typing
  - gate interval ownership
  - line interval ownership
  - design-time derivation
  - channel deduplication
  - center definition logic
  - definition assignment
  - type priority
  - authority priority
  - rejected old gate anchor assumptions
- Golden comparison is exact for pinned structure, identifiers, ordering, precision, and datetime formatting.

12 Delivery And Acceptance

- Required deliverables:
  - Trinity Engine implementation
  - deterministic execution entry point
  - pinned dependency specification
  - build and run instructions
  - golden test execution support
  - implementation notes for canon-sensitive areas
  - versioned release identifier
- Minimum acceptance conditions:

  - all required in-scope features implemented
  - no canon-critical ambiguity remains unresolved in delivered scope
  - required metadata present and correct
  - required golden tests pass
  - output matches canon exactly
  - no undisclosed canon deviation
  - no out-of-scope shortcuts simulating compliance
- Automatic rejection conditions include:
  - wrong gate or line assignment
  - wrong design-time method
  - wrong node policy
  - wrong channel, Type, or Authority derivation
  - wrong output contract
  - acceptance of invalid engine-boundary input

13 Canon-Control Items Worth Tracking During Implementation

- Document 09 item O02 says the consistency pass across Documents 05, 06, and 07 is resolved but not yet fully folded back.
- Document 10 is structurally defined but operationally incomplete until fixtures are created and frozen.
- Document 00 still treats final consistency, identifier hygiene, and fixture population as active risks.